

[Menu](#)

Database Administration

▼ Relevant source files

`admin/verification/checkXHTML.php``admin/verification/longOperationInit.php``admin/verification/orderLinksByTitle.php``admin/verification/rebuildCalculatedFields.php``admin/verification/rebuildRecordTitles.php``admin/verification/verifyFieldTypes.php``admin/verification/verifyValue.php``hclient/widgets/viewers/3dViewer.php``hclient/widgets/viewers/3dhopViewer.php`

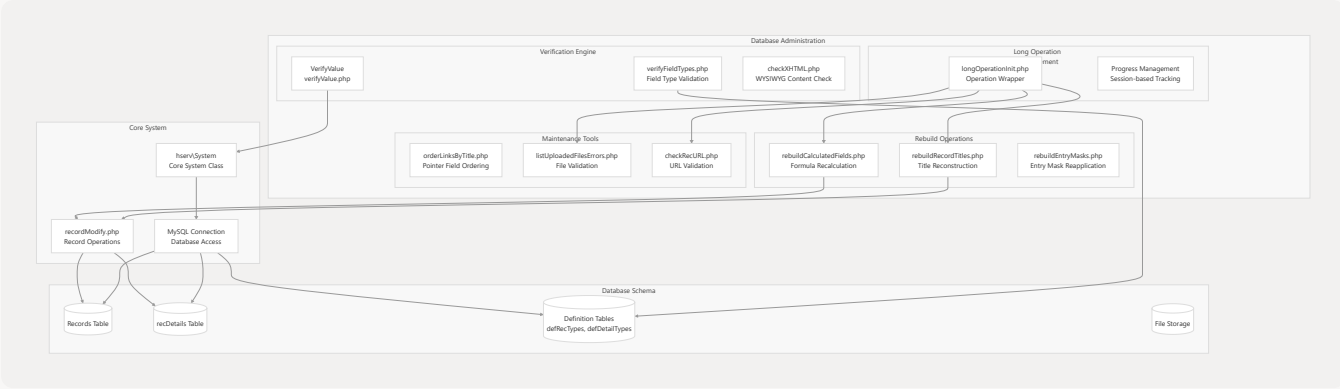
Purpose and Scope

The Database Administration system provides comprehensive tools for maintaining data integrity, verifying database consistency, and performing bulk maintenance operations in Heurist databases. This system encompasses verification utilities, rebuild operations, and long-running task management to ensure database health and performance.

For user management and permissions, see [User Management](#). For general system configuration, see [Configuration System](#).

Database Administration Architecture

The database administration system is organized around several core components that handle verification, maintenance, and bulk operations:



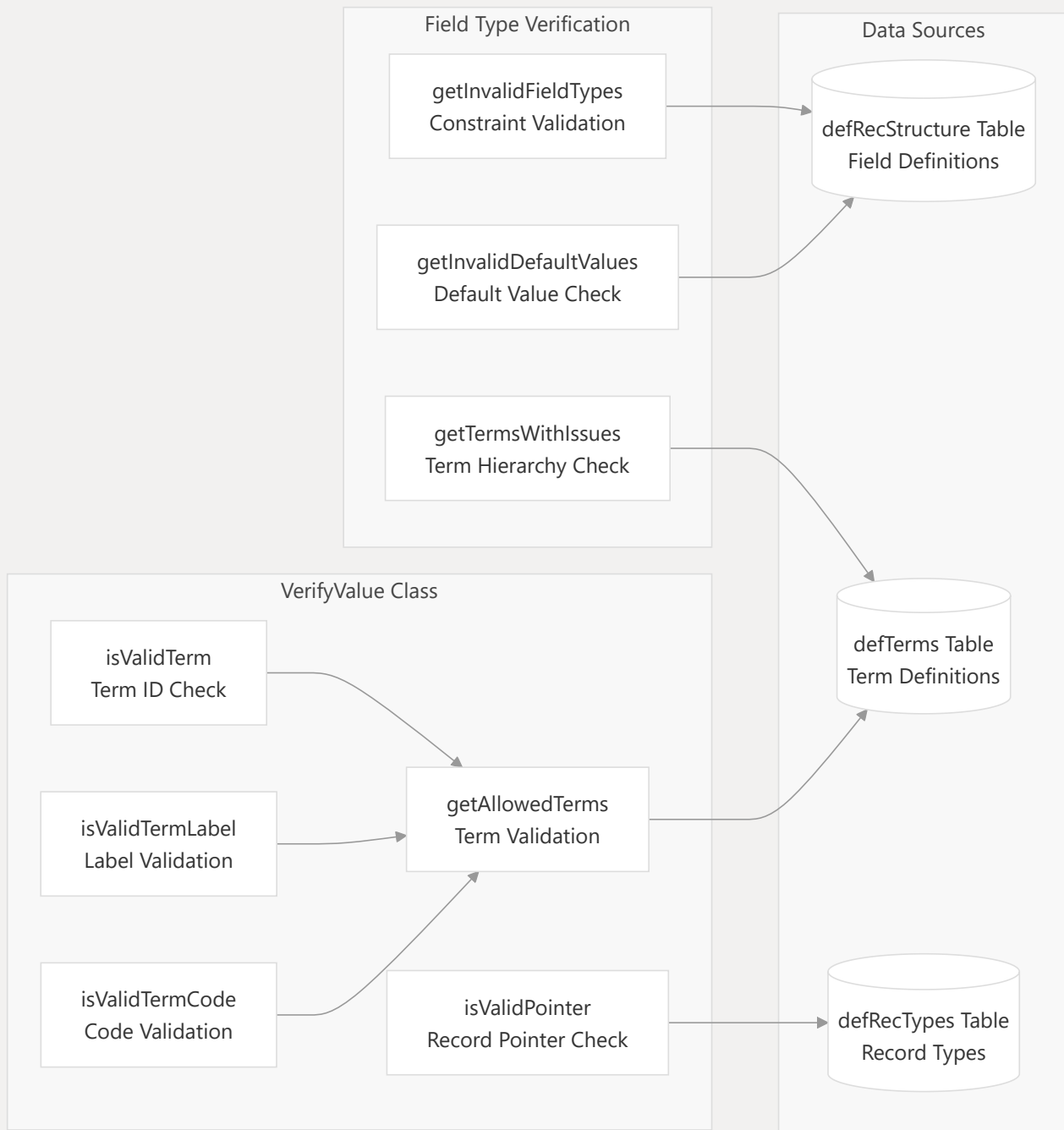
Sources: `admin/verification/longOperationInit.php` 1-112

`admin/verification/verifyValue.php` 1-296

`admin/verification/verifyFieldTypes.php` 1-403

Verification System

The verification system validates data integrity across multiple dimensions, from field-level constraints to cross-table relationships:



Term and Pointer Validation

The `VerifyValue` class provides static methods for validating data against schema constraints:

- **Term Validation:** `isValidTerm()` checks if term IDs conform to vocabulary constraints defined in `dtj_JsonTermIDTree`
- **Label Validation:** `isValidTermLabel()` validates hierarchical term labels with dot-separated notation
- **Pointer Validation:** `isValidPointer()` ensures record pointer fields reference valid records of allowed types

Sources: `admin/verification/verifyValue.php` 75-143

`admin/verification/verifyValue.php` 156-212 `admin/verification/verifyValue.php` 269-293

Field Type Integrity Checks

The field type verification system identifies inconsistencies in field definitions:

Check Type	Function	Purpose
Invalid Terms	<code>getInvalidFieldTypes()</code>	Validates <code>dtty_JsonTermIDTree</code> references
Non-selectable Terms	Same function	Checks <code>dtty_TermIDTreeNonSelectableIDs</code>
Record Type Constraints	Same function	Validates <code>dtty_PtrTargetRectypeIDs</code>
Default Values	<code>getInvalidDefaultValues()</code>	Verifies <code>rst_DefaultValue</code> integrity
Term Hierarchy	<code>getTermsWithIssues()</code>	Detects missing parents and duplicates

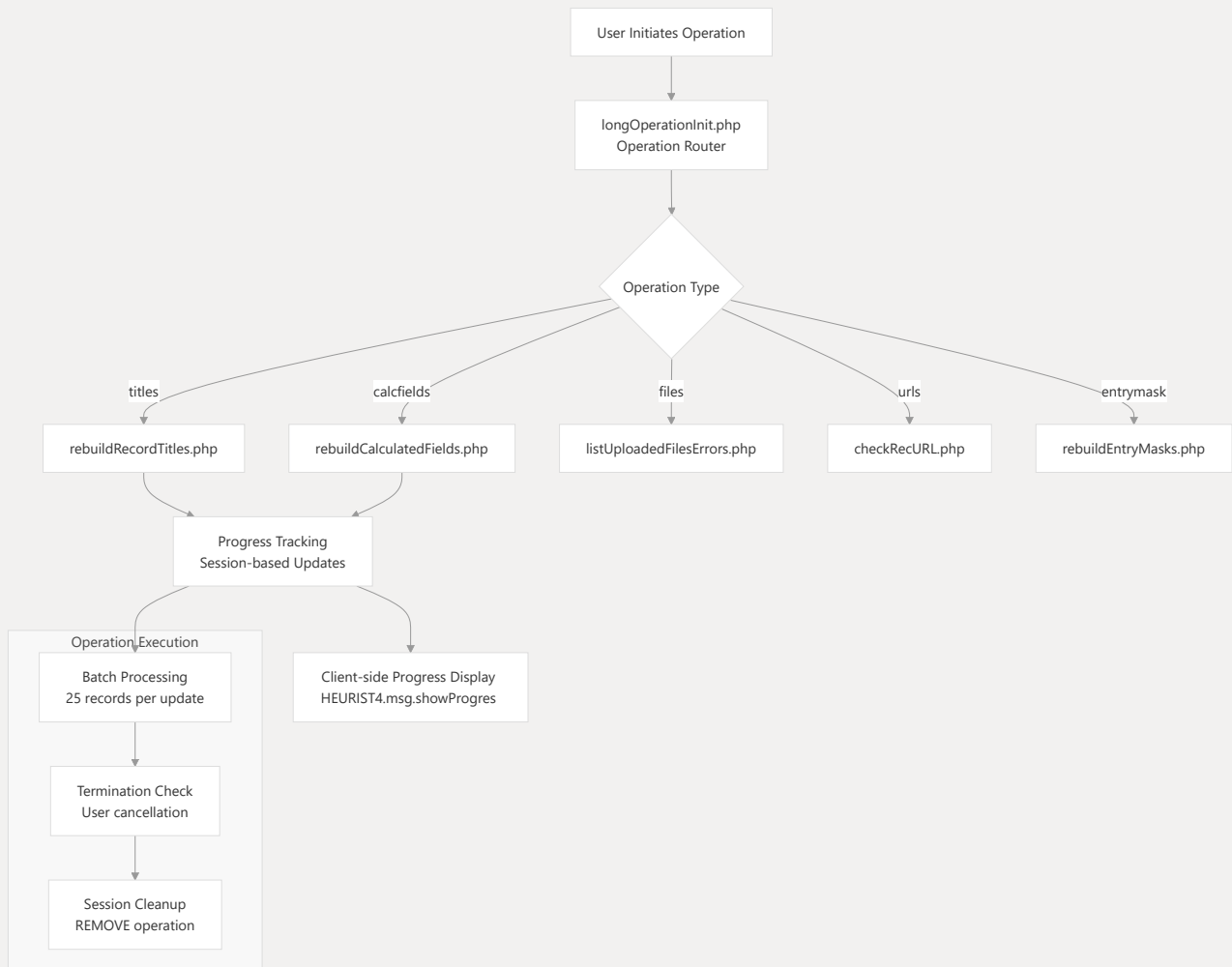
Sources: `admin/verification/verifyFieldTypes.php` 63-141

`admin/verification/verifyFieldTypes.php` 146-210

`admin/verification/verifyFieldTypes.php` 216-282

Long-Running Operations Management

The system handles time-intensive operations through a specialized wrapper that provides progress tracking and user feedback:



Progress Session Management

Long-running operations use session-based progress tracking implemented in the `progressSession()` function:

- **Initialization:** Creates progress session with total record count
- **Updates:** Reports progress every 25 processed records
- **Termination:** Checks for user cancellation requests
- **Cleanup:** Removes session files upon completion

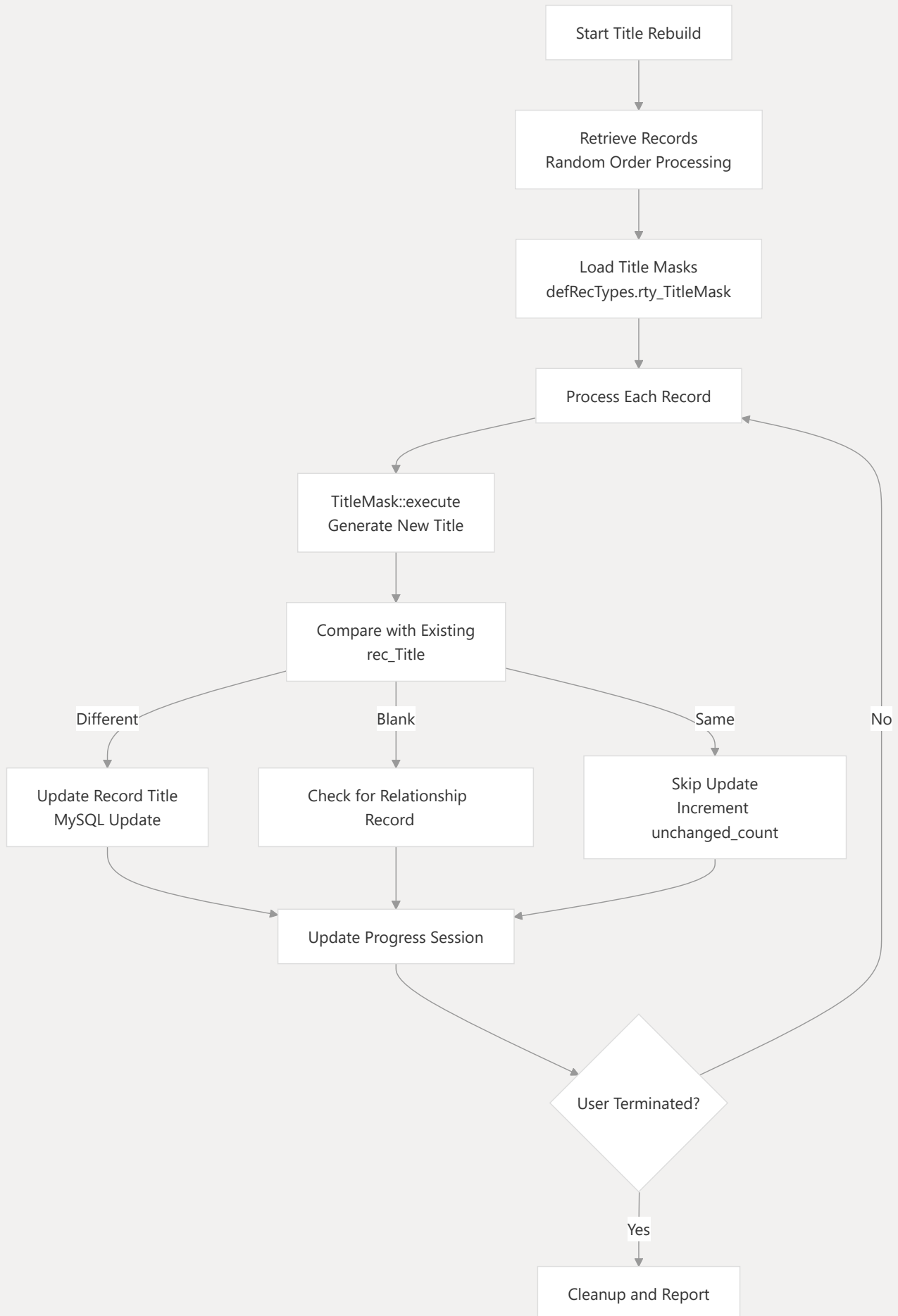
Sources: `admin/verification/rebuildRecordTitles.php` 328-348

`admin/verification/longOperationInit.php` 32-61

Rebuild Operations

Record Title Reconstruction

The `rebuildRecordTitles.php` system rebuilds constructed titles using title masks:



The rebuild process handles three outcomes:

- **Updated:** Title changed and updated in database
- **Unchanged:** Title identical to existing value
- **Blank:** New title would be empty (preserves existing title)

Sources: `admin/verification/rebuildRecordTitles.php` 234-324

`admin/verification/rebuildRecordTitles.php` 270-295

Calculated Fields Reconstruction

The `rebuildCalculatedFields.php` system recalculates formula-based fields using the `recordUpdateCalcFields()` function from the record modification system. This operation processes all calculated field types and reports changes, clearances, and errors.

Sources: `admin/verification/rebuildCalculatedFields.php` 43-75

File and Media Validation

The system includes specialized tools for validating file integrity and media references:

File Error Detection

The `listUploadedFilesErrors.php` tool (referenced but not provided in sources) validates:

- File existence in storage
- Metadata consistency
- MIME type accuracy
- File permissions and accessibility

URL Validation

The `checkRecURL.php` tool validates URL fields in records to identify broken links and connectivity issues.

3D Media Validation

The system includes viewers for 3D content validation:

- **3D Viewer:** Supports multiple formats (obj. 3ds. stl. plv. altf. alb) via `3dViewer.php`